



Short Communication

Search for spatial coincidence between IceCube neutrinos and gamma-ray bright red dwarfs

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A B S T R A C T

We search for a spatial coincidence between high energy neutrinos detected by the IceCube neutrino detector and ten red dwarfs which have been observed in gamma-rays. For our analysis, we use the unbinned maximum likelihood method to look for a statistically significant excess. We do not find any such spatial association between any of the red dwarfs and IceCube-detected neutrinos. Therefore, we conclude that none of the gamma-ray bright red dwarfs contribute to the diffuse neutrino flux measured by IceCube.

1. Introduction

The origin of majority of the IceCube diffuse neutrino flux in the TeV–PeV energy range (Aartsen et al., 2013) is still unknown (Halzen, 2023). Knowing the sources of the diffuse neutrino background flux will help us unravel a large number of problems in Physics and Astronomy from origin of ultra-high energy cosmic rays to beyond the standard model of particle physics (Argüelles et al., 2024). Although, searches by the IceCube collaboration have shown evidence for neutrino emission from some point sources such as NGC 1068, TXS 0506+056, NGC 4151, and PKS 1424+240, the majority of IceCube neutrinos cannot be attributed to any astrophysical sources (Abbasi et al., 2021). Searches for spatial coincidence with a plethora of extragalactic sources, which could contribute to the diffuse neutrino flux have been done. A non-exhaustive list of such extra-galactic sources includes AGNs and GRBs (Zhou et al., 2021; Hooper et al., 2019; Luo and Zhang, 2020; Smith et al., 2021; Li et al., 2022; Abbasi et al., 2011, 2022a; Aartsen et al., 2017; Abbasi et al., 2022b), FRBs (Luo and Zhang, 2024; Desai, 2023; Aartsen et al., 2020a), high energetic events from the Fermi-LAT catalog (Li et al., 2022), merging galaxies (Bouri et al., 2024), and galaxy catalogs using the 2MASS survey (Aartsen et al., 2020b) and WISE-2MASS sample (Zhou et al., 2024).

About a year ago, the IceCube collaboration found 4.5σ evidence for neutrino emission from the Galactic plane using cascade events (Abbasi et al., 2023). Although the observed signal is consistent with diffuse emission from the galactic plane, some of the signal could arise from a population of unresolved point sources such as supernova remnants,

pulsar wind nebulae or unidentified TeV Galactic sources (Abbasi et al., 2023).

In a similar vein, a galactic contribution to the all-sky IceCube diffuse neutrino flux cannot be excluded and has been estimated to be about 10–20% (Palladino and Vissani, 2016; Troitsky, 2021). Therefore, searches for coincidences with multiple galactic sources such as supernova remnants, X-ray binaries, pulsars, pulsar wind nebulae, LHAASO and HAWC sources, open clusters have also been carried out (Vance et al., 2021; Denton et al., 2017; Abbasi et al., 2022c; Aartsen et al., 2020c; Kovalev et al., 2022; Huang and Li, 2022; Li et al., 2022; Kumar et al., 2024; Pasumarti and Desai, 2022, 2024). A large number of galactic sources have also been targeted for neutrino astronomy by the previous generation of underground neutrino detectors (Ambrosio et al., 2001; Abe et al., 2006; Desai et al., 2008; Desai, 2022).

In this work, we look for high energy neutrinos from red dwarfs. Red dwarfs (sometimes also referred to as M-dwarfs) are one of the smallest stars on the main sequence with masses between $0.075\text{--}0.5 M_{\odot}$, having surface temperatures between 2,500–5,000 K (Edgeworth, 1946), and frequently emit flares with energies between 10^{32} and 10^{35} ergs with a power-law index between 1.7–2.4 (Yang et al., 2017). Energetic outbursts have been detected from red dwarfs at high energies from hard X-rays to TeV gamma-rays (Osten et al., 2016; Song and Paglione, 2020; Sinitsyna et al., 2019). Most recently, the SHALON TeV gamma-ray telescope detected very high energy gamma-ray emission between 800 GeV and 20 TeV from eight red dwarfs, namely, V388 Cas, V547 Cas, V780 Tau, V962 Tau, V1589 Cyg, GJ 1078, GJ 3684 and GL 851.1 (Sinitsyna et al., 2019). Subsequently, these red dwarfs have been proposed as sources of ultra high energy cosmic rays (Sinitsyna et

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al., 2022, 2021). However, no corresponding gamma-ray emission was seen at GeV (Huang et al., 2024) or MeV energies (Shrivastava et al., 2024). Song and Paglione (2020) reported a 5σ detection of pulsed GeV γ -ray emission from another M-dwarf TVLM 513-46546 with Fermi-LAT. A very hard X-ray outburst was detected by the SWIFT-BAT telescope from another M-dwarf, viz DG CVn from 0.3-100 keV in April 2014 (Osten et al., 2016), although no coincident gamma-ray emission was seen from Fermi-LAT (Loh et al., 2017). Although there are no estimates for high energy neutrino flux emission from red dwarfs, these red dwarfs could generate cosmic rays with energies up to 10^{14} eV in their flaring states (Sinitsyna et al., 2021). TeV gamma-rays detected from these red dwarfs indicate that these red dwarfs produce flares with total energies of $10^{32} - 10^{35}$ ergs (Sinitsyna et al., 2022, 2021). If one considers the statistics of M-dwarfs in our galaxy and assuming the frequency of flares to be around 36/year for about 10% of the observed red dwarfs, the total energy released from these flares matches the total energy of cosmic rays in the galactic disk of $\sim 10^{54}$ ergs (Sinitsyna et al., 2019). During these flares the accelerated protons with energies up to 10^{14} eV undergo spallation with the atmosphere outside the star (Sinitsyna et al., 2021). Note that similar mechanisms for production of TeV gamma-rays have also been proposed for the Sun (Zhou et al., 2017). As argued in Sinitsyna et al. (2021), these interactions produce pions. Consequently, one would also expect neutrino emission from the aforementioned hadronic interactions along with gamma-rays produced from pion decay. Therefore, these considerations provide enough motivation to search for neutrino emissions to complement recent searches for MeV (Shrivastava et al., 2024) and GeV (Huang et al., 2024) gamma-rays from these red dwarfs.

Therefore, in this work we look for neutrino emission from the aforementioned 10 red dwarfs using the publicly available muon track data from IceCube. We have previously used this data to look for neutrino emission from pulsars and FRBs (Desai, 2023; Pasumarti and Desai, 2022, 2024). The outline of this manuscript is as follows. The neutrino dataset used for the analysis is discussed in Section 2. The analysis and results are discussed in Section 3. We conclude in Section 4.

2. Dataset

The neutrinos used in this work are taken from the IceCube 10 year muon track data (Abbasi et al., 2021). IceCube is a detector located at the South Pole, which detects neutrinos through the Cherenkov radiation emitted by the charged leptons traveling through the ice.

This dataset consists of 1,134,431 track-like neutrino candidates, which were collected between April 2008 (IC-40) and July 2018 (IC 86-V11) from four different phases of the experiment, each having different livetime. Note that for our analysis we have used the augmented dataset analyzed in Zhou and Beacom (2022),¹ which corrects for 38 duplicate events found in the IceCube public muon track data due to angular reconstruction error (Zhou and Beacom, 2022). For each neutrino, the dataset consists of right ascension (RA), declination (δ), error in the position, and reconstructed muon energy.

3. Analysis and results

For our analysis, we are using the unbinned maximum likelihood ratio method (see Pasumarti and Desai, 2022 and references therein, and first proposed in Braun et al., 2008). We use neutrinos within a declination of $\pm 5^\circ$ of the red dwarf. For a dataset containing N events, if n_s is the number of signals attributed to the red dwarf, then the likelihood of the entire dataset is given by:

$$\mathcal{L}(n_s) = \prod_{i=1}^N \left[\frac{n_s}{N} S_i + \left(1 - \frac{n_s}{N}\right) B_i \right] \quad (1)$$

¹ This dataset is available at https://github.com/beizhouphys/IceCube_data_2008-2018_double_counting_corrected.

Table 1

Results from spatial coincidence analysis between IceCube neutrinos and red dwarfs. None of the red dwarfs shows a detection significance $> 5\sigma$, corresponding to $TS > 25$. Therefore, we do not find any spatial association between the red dwarfs and IceCube neutrinos. The last column shows the 95% c.l. upper limit on the number of signal neutrino events.

Red dwarf	RA ($^\circ$)	δ ($^\circ$)	$\hat{n}_s$	TS	Upper limit
V388 Cas	15.8326	+62.365	0.0	0.0	16.2
V547 Cas	8.1229	+67.235	0.0	0.0	16.7
V780 Tau	85.107	+24.802	6.00	0.35	29.8
V962 Tau	86.466	+22.879	0.0	0.0	13.8
V1589 Cyg	310.705	+41.38	0.0	0.0	12.7
GJ 1078	80.95	+22.54	20.39	2.38	51.2
GJ 3684	176.77	+70.03	0.0	0.0	14.9
GL 851.1	333.0267	+31.56	0.0	0.0	11.1
TVLM 513-46546	225.284	22.834	0.0	0.0	20.8
DG CVn	202.944	29.28	2.42	0.05	27.2

where S_i is the signal PDF and B_i is the background PDF. Here the signal PDF is given by:

$$S_i = \frac{1}{2\pi\sigma_i^2} e^{-(|\theta_i - \theta_s|)^2 / 2\sigma_i^2} \quad (2)$$

where $|\theta_i - \theta_s|$ is the angular distance between the neutrino and the red dwarf, σ_i is the angular uncertainty in the neutrino position, in radians. The background PDF can be obtained from the solid angle within the declination (δ) of $\pm 5^\circ$ around each red dwarf and is given as follows Braun et al. (2008):

$$B_i = \frac{1}{\Omega_{\delta \pm 5^\circ}} \quad (3)$$

We then evaluate $\mathcal{L}(n_s)$ for different values of n_s , and designate the value of n_s , for which this likelihood is maximized as $\hat{n}_s$. Now to ascertain the significance of the signal, we define the test statistics (TS) based on the likelihood ratio, with respect to the null hypothesis in the absence of a signal, ($\mathcal{L}(0)$) as follows:

$$TS = 2 \log \frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(0)}. \quad (4)$$

For the null hypothesis, TS behaves like a χ^2 distribution for one degree of freedom Wilks (1938). The detection significance can be obtained from $\sqrt{TS}$. This statistics is also used in gamma-ray astronomy (e.g. Manna and Desai, 2024). For a statistically significant detection of $> 5\sigma$, $TS > 25$. For each of our red dwarfs, we determine $\hat{n}_s$ and evaluate TS according to Eq. (4). These values of TS , along with the values of $\hat{n}_s$ can be found in Table 1. As we can see, none of the red dwarfs show a TS value > 25 . The largest TS value is seen for GJ 1078 with TS of 2.38, whose significance is less than 2σ . Therefore, the observed signal events are consistent with background, and there is no evidence for any spatial association between red dwarfs and IceCube neutrinos, which are observed as muon tracks. Consequently, we calculate the 95% c.l. upper flux limits by calculating the value of n_s for which $\Delta(TS) = -3.84$ Zyla et al. (2020), where $\Delta TS = TS - TS_{max}$.

4. Conclusions

In this work, we searched for a spatial coincidence between IceCube neutrinos and 10 gamma-ray bright red dwarfs. The neutrinos used in this work are taken from the publicly released 10 year muon track data observed between 2008-2018, and consisting of 113443 neutrinos. Eight of the 10 red dwarfs were detected at TeV gamma rays by the SHALON telescope. Two others were detected in gamma-rays and X-rays from Fermi-LAT and SWIFT. We analyzed these data using the unbinned maximum likelihood method. A tabular summary of our results for all the red dwarfs can be found in Table 1. All the red dwarfs show a test statistic value of less than two, with significance less than 2σ . Therefore, we

conclude that none of the above red dwarfs contribute to the Ice Cube diffuse neutrino flux.

CRediT authorship contribution statement

Fathima Shifa M: Formal analysis. **Shantanu Desai:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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